

JASON CUSHEN

Computer Systems Graduate · Full-Stack Development · Software Engineering

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PROFILE

Final-year Computer Systems student at the University of Limerick with hands-on industry experience gained at Jaguar Land Rover and a strong foundation in full-stack web development. Built a production-quality, end-to-end Gaelic football analysis platform as a Final Year Project, covering system architecture, database design, cloud media integration, and data visualisation. Comfortable working across the full stack, from REST API design and backend data pipelines to React-based frontend interfaces. Experienced in agile development environments, collaborative team working, and delivering reliable, well-tested software.

EXPERIENCE

Undergraduate Software Developer · Jaguar Land Rover (JLR)

May 2024 – Jan 2025

Shannon Software Engineering Centre, Co. Clare, Ireland

- ▶ Worked within JLR's Test Platform department, whose core purpose was building and maintaining automated test infrastructure for vehicle software and hardware validation across global engineering sites.
- ▶ Embedded in three agile squads over eight months, contributing to daily stand-ups, sprint planning, backlog refinement, and retrospectives across each rotation.
- ▶ Contributed to an internal platform used by test engineers to schedule and execute keyword-driven test cases on physical and virtual test assets located across multiple global sites, including Hardware-in-the-Loop (HIL) rigs running on dSPACE, gaining direct exposure to automotive ECU validation pipelines and how hardware test infrastructure is structured and operated at scale.
- ▶ Worked on the asset management system responsible for tracking the availability, status, and full lifecycle of over 900 physical and virtual test assets across the organisation.
- ▶ Validated developed code by executing test cases against virtual workbenches within the test platform, gaining familiarity with the software-to-hardware testing pipeline used across JLR's engineering operations.
- ▶ Documented and maintained test cases, test plans, and traceability links within Jira and XRay, and integrated requirements into the 3DXperience platform using the Traceable Requirements Management toolset.
- ▶ Developed Python automation scripts and Groovy scripts to automate workflow stages within the test asset management system, reducing manual overhead in tracking physical test assets through their lifecycle.
- ▶ Participated in the company-wide undergraduate challenge, collaborating with other undergraduates within the company to design, prototype, and present a driver-facing camera system for vehicle theft prevention; explored OpenCV and Python computer vision libraries and presented findings to a panel of company judges.

General Operative · Altrad Services · Moneypoint Power Station

Summer 2023 & 2025

Moneypoint, Co. Clare, Ireland

- ▶ Worked for Altrad within Moneypoint Power Station as a general operative, undertaking a range of duties including general labour, scaffolding support, and routine maintenance. Gained experience working within a large industrial workforce, adhering to strict health and safety protocols and consistently meeting daily operational demands."

PROJECTS

Web-Based Gaelic Football Analysis Tool

Final Year Project · 2025 / 2026

University of Limerick · Supervisor: Prof. Alison O'Connor

- ▶ Designed and built a full-stack web application from the ground up to address a clear gap in the GAA community: the absence of an accessible, purpose-built performance analysis tool for Gaelic football.
- ▶ Built the presentation layer as a Next.js application with a fully component-based React UI and TypeScript throughout; implemented authentication, team and player management, and a dashboard routing structure.
- ▶ Designed and implemented all core features: video upload and playback, interactive event tagging on a coordinate-mapped pitch, automatic running scoreboard, and a clip extraction system allowing users to generate short highlight clips around any tagged event.
- ▶ Integrated Cloudflare Stream for scalable match video hosting and clip generation via REST API, and Cloudflare R2 for PDF report storage; implemented server-side signed URL access to prevent unauthorised direct access to media assets.
- ▶ Used Supabase for all structured data persistence, implementing Row-Level Security policies to enforce per-user data isolation at the database level, with a service-role bypass pattern for validated cross-user access in group workspaces.
- ▶ Built a Python/LaTeX PDF report generation pipeline invoked server-side from a Next.js API route, computing match statistics (kickout percentages, shot conversion rates, player-level scoring, turnover counts) and producing structured downloadable reports.
- ▶ Implemented a group collaboration layer: group workspaces with invite codes, admin-controlled join requests, and a messaging system supporting report and clip sharing between team members, addressing a key pain point identified in interviews.

CSIS Equipment Booking System

University Hackathon · 2024

University of Limerick · 2-Day Event

- ▶ Collaborated in a team to design and develop a full-stack equipment booking system for the Computer Science department within a 2-day hackathon format, covering full design, development, and delivery under strict time constraints.

EDUCATION

B.Sc. Computer Systems · University of Limerick

Sept 2022 – Sept 2026

Limerick, Ireland

- ▶ Current QCA: 3.02 (final results pending).
- ▶ Relevant coursework: Data Structures & Algorithms, Operating Systems, Databases, Software Engineering, Machine Learning.
- ▶ Cooperative Education placement with Jaguar Land Rover (8 months) — integrated as part of the degree programme.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript

Frontend: React, Next.js, CSS, Tailwind CSS

Backend & APIs: REST API design, JWT authentication

Databases & Storage: Supabase, PostgreSQL, SQL, Cloudflare

Dev Tools: Git, GitHub, GitLab, Jira, XRay, Confluence, Agile / Scrum

Other: Python automation, Groovy scripting, OpenCV, LaTeX, pdflatex
